Python Task 25-06-26

1. **Integer:**An Integer is a data type which stores whole numbers including positive numbers negative numbers and zeros without decimals.
   Features:1.immutabe
   2. No slicing allowed
   3.Length cannot be found
   Ex: a=16
   print(a)
   print(type(a))
   o/p: 16
   <class ‘int’>

2.**String:**A string is sequence of characters used to store text data and enclosed between single double or triple quotes.
Features:1.Immutable
2.Supports indexing and slicing
3.String length can be found
4.Orderd
5. Iterable
Ex:
a=”sivani”
print(a)
print(type(a))
print(a(2))
print(a[3:])
o/p:sivani
<class ‘str’>
v
ani

3.**List:**A list is a collection of different data type elements enclosed between [ ].
Features:1.Allows heterogenous objects
2.Mutable
3.Ordered
4.Supports indexing and slicing
5.Iterable
6.Allows duplicate values
Ex: names = ["Sivani", "Suchitra", "Akshara",4,7.5,4,True]
names[1] = "Anjali"
print(names[4])
print(names[1:3])
o/p:[‘Sivani’,’Anjali’,’Akshara’4,7.5,4,True]
7.5
[‘Anjali’,’Akshara’]

4.**Tuple:**A tuple is a collection of elements enclosed between ( ).
Features:1.Immutable
2.Allows heterogenous values
3.Ordered
4.Supports indexing and slicing
5.Iterable

Ex:
```
n=(10,20,30,40,50)
Print(n[3])
print(n[2:5])
For i in n:
    print(i)
```
o/p:
```
40
(30,40,50)
10
20
30
40
50
60
```

5.**Set:** A set is an unorderd collection of unique elements enclosed by { }.
Features:1.Mutable
2. duplicate values are not allowed
3.unordered
4.Does not supports indexing and slicing

Ex:
```
numbers = {10, 20, 30, 40, 20, 50}
print(numbers)
numbers.add(60)
print(numbers)
for i in numbers:
print(i)
```
o/p:
```
{10,20,30,40,50}
{10,20,30,40,50,60}\
10
20
30
40
50
60
```

6.**Dictonary:**A dictionary stores key value paris eclosed between { }.
Features:1.Mutable
2.Keys must be unique
3.Duplicate values are not allowed

4.Data can be accessed using keys

Ex:student = { "id": 102, "name": "Sivani", "course": "Python", "marks": 55}
print(student)
print(type(student))
print(student["name"])
student["marks"] = 78
print(student)

o/p:{'id': 101, 'name': 'Sivani', 'course': 'Python', 'marks': 55}
<class 'dict'>
Sivani
{'id': 101, 'name': 'Sivani', 'course': 'Python', 'marks': 78}

7.**Mutable:** List,Set,Dicionary
Immutable: integer, String, tuple

8.The data types that allows duplicate values are : String,List,Tuple,Dictionary(only values can be duplicate).

9.**Insertion order:** String,List,tuple,Dictionary

10.The data types that supports indexing and slicing are : String,List,Tuple.

11.
```
a=4
b=”Sivani”
c=[1,2,3]
d=(4,5,6)
e={65,45,62}
f={“Fruit”:”grapes”,”vegetable”:”Potato”}
g=7.8
h=True
i=3+6j

print(type(a))
print(type(b))
print(type(c))
print(type(d))
print(type(e))
print(type(f))
print(type(g))
print(type(h))
print(type(i))
```

o/p:<class ‘int’>
<class ‘str’>
<class ‘list’>
<class ‘tuple’>

<class ‘set’>
<class ‘dict’>
<class ‘Float’’>
<class ‘Bool’>
<class ‘complex’>

12.

Data type comparision table

Features	Stores	Mutable	ordered	Duplicates	indexing	slicing
Integer (10)	whole numbers	NO	NO	not Allowed	NO	NO
String "ab"	characters	NO	yes	yes	yes	yes
List [ ]	collection	yes	yes	yes	yes	yes
Tuple ( )	collection	NO	yes	yes	yes	yes
Set { }	collection	yes	NO	NO	NO	NO
Dictionary {key:value}	key value pairs	yes	yes	only values can be duplicate	NO	NO